

The University of Lahore, CS & IT Department

Object Oriented Programming

Assignment # 02

Start Date: 07/11/2024

Total Marks: 20

Due Date: 01/12/2024

Program: BSCS

Instructions

1. Understanding the problem is part of the assignment. So, no query, please.
 2. You will get zero marks if found any type of plagiarism or AI generated code.
 3. No submission after due date.
 4. Upload pdf file containing your code and screenshots of outputs.
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Q1: University of Lahore Management System

The University of Lahore wants a simple program to manage essential information about its departments, teachers, students, and courses. The goal is to design a menu-driven program in C++ that can simulate basic university management functions.

Requirements:

You need to create a program that enables users to:

1. **Create Departments:** Allow the user to add new departments to the university. Each department should have a unique name.
2. **Manage Department Entities:**
 - **Add Teachers:** Each department can have multiple teachers, with information such as name, age, degree, and years of experience.
 - **Add Students:** Each department can enroll students, with details like name, age, and a unique student ID.
 - **Add Courses:** Each department offers courses with details such as course name and course code.
3. **Display Information:** The program should allow the user to display:
 - All departments within the university.
 - All teachers, students, and courses within a specific department.

Implementation Details:

1. **Person Class:** Design a Person base class to store common attributes like name and age for students and teachers.
2. **Inheritance** and Composition:

- **Teacher Class:** Use inheritance for the Teacher class, which inherits from Person. Additionally, use a Qualification class (composition) to store a teacher's degree and years of experience.
 - **Student Class:** Inherit the Person class for students and add a unique student ID.
3. **Course Class:** Create a Course class with attributes like course name and course code.
 4. **Department Class:** Design a Department class to manage lists of teachers, students, and courses, with a maximum of 10 entries per type.
 5. **University Class:** The University class should manage departments within the university, allowing the user to add departments and view department-specific details through a main menu.

Functional Requirements:

The program should be fully menu-driven, with options allowing the user to:

- Add new departments.
- Enter each department to manage teachers, students, and courses.
- Display all departments and the details of each teacher, student, and course within a department.

Constraints:

- Limit the number of teachers, students, and courses to 10 per department.
- The system should work only through menu options, allowing the user to navigate by selecting options to add or view data.

Note: Sample output is given bellow.

Output - 1

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 2

No departments available.

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 1

Enter department name: Computer Science

Department added successfully!

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 1

Enter department name: Software Engineering

Department added successfully!

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 2

1. Computer Science
2. Software Engineering

Output - 2

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 3

1. Computer Science
2. Software Engineering

Enter department number to manage: 1

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 4

No teachers available.

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 1

Enter details for teacher #1:

Enter name: Ali Ahmad

Enter age: 28

Enter degree: MS Computer Science

Enter years of experience: 4

Teacher added successfully!

Output - 3

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 1

Enter details for teacher #2:

Enter name: Noman Khalid

Enter age: 33

Enter degree: PhD Computer Science

Enter years of experience: 3

Teacher added successfully!

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 2

Enter details for student #1:

Enter name: Kashif Farooq

Enter age: 22

Enter student ID: 7019383

Student added successfully!

Output - 4

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 3

Enter details for course #1:

Enter course name: OOP

Enter course code: 201

Course added successfully!

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 4

Teachers in Computer Science Department:

Teacher #1:

Name: Ali Ahmad, Age: 28

Degree: MS Computer Science, Years of Experience: 4

Teacher #2:

Name: Noman Khalid, Age: 33

Degree: PhD Computer Science, Years of Experience: 3

Output - 5

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 5

Students in Computer Science Department:

Student #1:

Student ID: 7019383, Name: Kashif Farooq, Age: 22

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 6

Courses in Computer Science Department:

Course #1:

Course: OOP, Code: 201

--- Computer Science Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 0

Going back to main menu

--- University of Lahore Main Menu ---

1. Add Department
2. Show Departments
3. Enter Department
0. Exit

Enter your choice: 3

1. Computer Science
2. Software Engineering

Enter department number to manage: 2

--- Software Engineering Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 1

Enter details for teacher #1:

Enter name: Muhammad Hammad

Enter age: 29

Enter degree: MS Software Engineering

Enter years of experience: 2

Teacher added successfully!

--- Software Engineering Department Menu ---

1. Add Teacher
2. Add Student
3. Add Course
4. Display All Teachers
5. Display All Students
6. Display All Courses
0. Go Back

Enter your choice: 4

Teachers in Software Engineering Department:

Teacher #1:

Name: Muhammad Hammad, Age: 29

Degree: MS Software Engineering, Years of Experience: 2